

Nature of Energy

CONCEPT 1 NOTES

Energy

- Energy: the ability to cause change.
 - Measured in Joules (J)
 - 1 Joule = 1 kg*m²/s²
 - *There are many different types of energy. We will start by learning about kinetic and potential.*



Kinetic Energy

- Kinetic Energy: (KE) energy in motion.

$$KE = \frac{1}{2} mv^2$$

- Mass measured in (kg)
- Velocity measured in (m/s)



Example #1

A jogger with a mass of 60 kg is moving at a speed of 3.0 m/s. What is the jogger's kinetic energy?

$$KE = ?$$

$$m = 60.0 \text{ kg}$$

$$v = 3.0 \text{ m/s}$$

$$KE = \frac{1}{2} mv^2$$

$$KE = \frac{1}{2} (60.0)(3.0)^2$$

$$KE = (30)(9)$$

$$\mathbf{KE = 270 \text{ J}}$$

Practice Time

1. A baseball with a mass of 0.15 kg is moving at a speed of 40 m/s. What is the baseball's kinetic energy?

Answer: $KE = 120 \text{ J}$

2. If a stone has 390 J of energy and is moving with a speed of 12 m/s, what is the mass of the stone?

Answer: $m = 5.42 \text{ kg}$

3. What is the velocity of a 0.06 kg ball moving with 75 J of energy?

Answer: $v = 50 \text{ m/s}$

Potential Energy

- Potential Energy:
stored energy
 - *3 types of potential energy:*
 - Elastic potential energy
 - Chemical potential energy
 - Gravitational potential energy

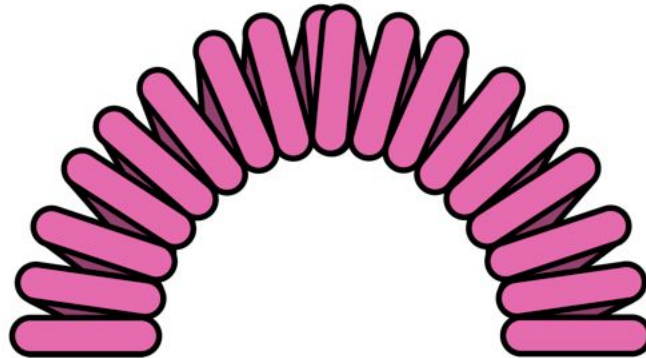
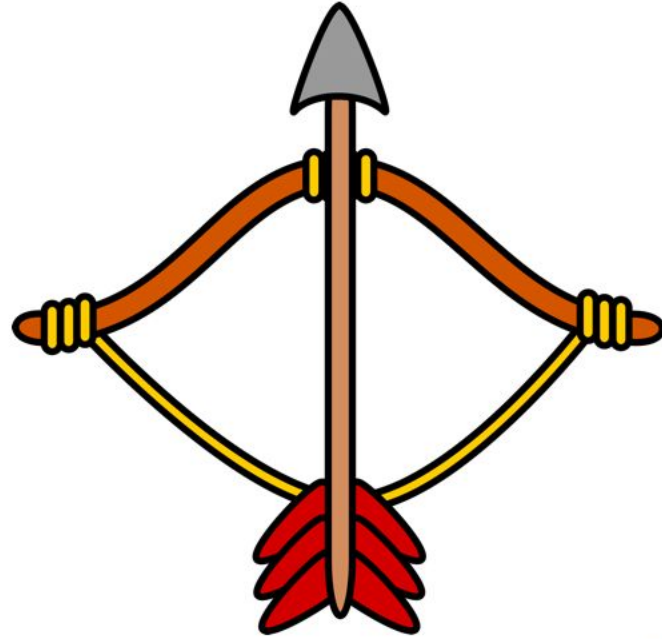


Potential Energy

- Elastic Potential Energy: energy stored in objects being compressed or stretched

– *Examples:*

- Bow and arrow
- Spring
- Slingshot

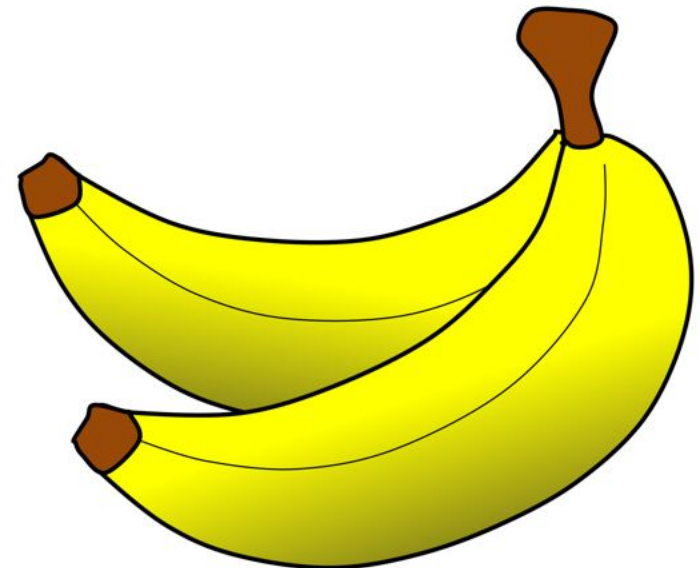


Potential Energy

- Chemical
Potential Energy:
energy stored in
chemical bonds of
compounds.

– *Examples:*

- Firewood
- Battery
- Food

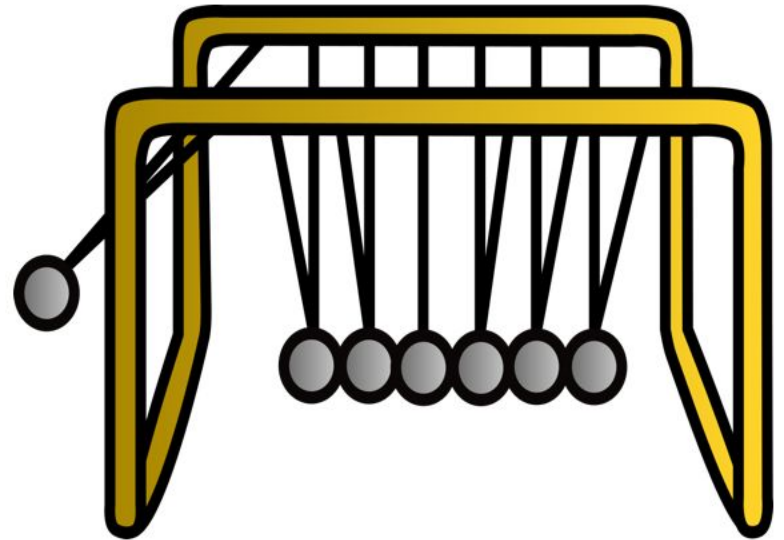


Potential Energy

- Gravitational Potential Energy: (GPE) energy stored in objects that are above Earth's surface.

$$\text{GPE} = \text{ham}$$

- **Height** measured in (m)
- **Acceleration** due to gravity measured in (m/s^2)
- **Mass** measured in (kg)



At the peak of its path, the sphere pauses for just a second and has gravitational potential energy, before it converts back to kinetic energy as it moves towards the other spheres.

Example #2

How much gravitational potential energy does a 5 kg rock have if it is sitting on the edge of a cliff that is 10 m high?

$$\text{GPE} = ?$$

$$m = 5 \text{ kg}$$

$$a = 9.8 \text{ m/s}^2$$

$$h = 10 \text{ m}$$

$$\text{GPE} = ham$$

$$\text{GPE} = ham$$

$$\text{GPE} = (10)(9.8)(5)$$

$$\text{GPE} = 490 \text{ J}$$

Practice Time

1. A Viking ship roller coaster at the fair has a mass of 36,000 kg. If at its peak it reaches a height of 20 m off the ground, how much gravitational potential energy does it have?

Answer: GPE = **7,056,000 J**

2. A 1.5 kg apple pie is sitting on the kitchen counter. If it has 13.44 J of gravitational potential energy, how tall is the counter it sits on?

Answer: h = **0.914 m**

3. A suitcase weighing 120 N sits on a counter 1.8 m high in the airport. How much energy does it have?

Answer: GPE = **216 J**

Practice Time